# Best Practices for Designing Semantic Models for AI Readiness

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# CONTENTS

# Overview

Designing semantic models that are AI-ready is no longer optional – it's critical for enabling agents to generate accurate, reliable, and consistent answers. Based on our AI Readiness Assessment initiative, here are the core best practices to maximize your  potential.

It's important to highlight that this document focuses specifically on AI readiness standards and best practices, which build upon and extend the well-established industry best practices for data modeling and data quality.

# Supported Semantic Model Features

To guarantee compatibility with Agentforce for Analytics, make sure your semantic model adheres to the following:

1. **DMOs Only:** The semantic model should contain Data Model Objects (DMOs). Data Lake Objects (DLOs) and Calculated Insights (CIs) aren't currently supported.
2. **Full Relationship Map**: The semantic model must contain a complete and accurate relationship map of your data.

# Best Practices for Semantic Model AI Readiness

## Define Clear Object and Field Names

Labels = AKA Display Name, Object/Field Name, Name.

Agents interpret fields in the context of the objects they belong to. Therefore, the clarity and specificity of both object names and field labels play a crucial role in AI interpretability. While generic label names like "ID" or "Name" seem ambiguous in isolation, they're acceptable if the object they belong to is well-named and clearly described. It's the combination of object context and field properties that drives semantic clarity.

In addition, both `Api_Name` (the system name defined at creation in Data Cloud) and `Name` (the display label) are exposed to AI agents. While the `Api_Name` can't be edited postcreation in the semantic model, it should be created thoughtfully. The `Name` can and should be tailored for business clarity.

> **Best Practice**: Ensure the combination of object name, description, and field label is sufficient for uniquely identifying the field's role and purpose.
>
> **Best Practice:** Use meaningful, business-relevant display names, especially for custom fields. For example, pair an object named "Customer" with a field labeled "ID" – this is more interpretable than an ambiguous pairing like "Entity" with "ID".

> **Best Practice:** When creating new fields in Data Cloud, choose `Api_Names` that are as descriptive as possible to improve downstream interpretability.

## Provide Informative, Balanced Descriptions

Descriptions play a pivotal role in helping agents interpret fields and objects accurately. For Salesforce Standard Objects, leaving the default description may be acceptable, as these are widely known and referenced across public knowledge sources that LLMs are trained on. However, if a user modifies a standard description, it must be additive and not contradict existing knowledge.

> **Best Practice:** Use Salesforce Standard Objects and Fields when designing your data model. LLMs are more likely to understand and interpret standard components correctly because they are documented more and are used in public training data.
>
> **Best Practice:** Before updating a standard field description, validate its existing meaning by querying your preferred LLM chat tool. Make sure your update enhances or aligns with known usage.

For Custom Objects and Fields, a clear and informative description is essential. It should reflect what the data represents, align with industry standards whenever possible, and explain any unique internal meanings when necessary.

> **Best Practice**: Provide a concise, 1–2 sentence description that:
>
> 1. Conveys the business context (for example "Purchase Date represents the Date field for when the invoice was printed or sent to the user. In some cases the Transaction Date and Purchase Date are different due to that reason.")
> 2. Describes how it's being used in the business (for example "Purchase Date is the default Date filter and slicer in most Purchasing and Costing related analysis.")
> 3. Represents the data well. Review the data, and make sure the description doesn't only describe the general purpose of the field. It should elaborate if there are specific data values with a specific purpose. (for example "NA" Represents transactions in which their purchase wasn't confirmed yet. Which aligns with "Transaction Status" = 'Pending')
>
> Once done, test it against LLM understanding by asking your favorite chat tool "How do you interpret this [Your Description Content] for the field [Your Field Name]?".
>
> **Best Practice:** Provide a clear description of the semantic model that conveys its goals and use cases. It provides essential context about the model's purpose and scope, and improves the agent's ability to reason about its content.

# Detect and Resolve Ambiguities

Agents commonly analyze the full component properties when interpreting semantic models. However, it's essential to ensure that any ambiguity can be resolved solely based on the label and description level. Ambiguity can arise from the existence of similar entities (objects), similar fields, calculated fields, and metrics – and sometimes even between calculated fields and regular fields.

Agentforce Analytics Agent evaluates the complete set of object properties–including Object Name, Object Description, Field Display Name, Field Descriptions, Field Data Type, Field Role, Sample Values, and object/field relationships–to reduce ambiguity. However, because it's common for multiple fields to carry similar data or serve overlapping purposes, the semantic model must clearly differentiate between them through precise and intentional metadata.

> **Best Practice:** Regularly run similarity and ambiguity checks. Ensure that each component created and added to the semantic model serves a unique purpose, and that this purpose is clearly captured in both the label and description.

## Common Sources of Ambiguity

1. Overlapping Descriptions – Vague or identical field descriptions that fail to clarify distinct roles or usage. Since it's common to copy and paste descriptions, often it leads to ambiguity.
2. Similar Business Context – Fields that represent the same concept but appear in different tables (for example “Customer Region” from Orders Header vs. Order Line).
3. Synonyms or Homonyms – Fields with different names representing the same concept (synonyms), or fields with identical names representing different concepts (homonyms).
4. Objects which exist twice for different purposes and uses, but it's not clear from their semantics that they differ. Here are some common examples:
    a. Same object, different purpose - Generic Date dimension in the Datawarehouse, which is being used once with Purchase_Date, and second for Transaction_Date
    b. Same object, different granularity - Order Lines table used for drill down to the order level when needed, and an aggregated version of Order Lines to the Date level granularity created for performance optimization.
    c. The user used a base model that has 10 objects. One of the objects is an Account object that has only five of the fields they need. Due to this constraint and in order to not adjust the base model, an additional Account object was added with all columns (existing and new). Similar to Extended models, these cases may be encountered in models with PDS (Published Data Sources) and Logical Views.

# Manage Similar Calculated Fields

Agents frequently analyze formula definitions directly, which can introduce confusion even when labels and descriptions are clear. Calculated fields with similar or overlapping logic can lead to misinterpretation of the semantic model.

Similarly, metrics that differ only by minor filter variations can create comparable confusion. These cases should be managed with the same rigor applied to calculated fields.

> **Best Practice:** Avoid having similar calculations. If necessary, hide or remove calculated fields and metrics that don't deliver a clear, distinct analytical purpose. If removal isn't feasible, clearly define their role and differentiate them explicitly in their label and description to avoid ambiguity at the formula level.

# Enriching the Semantic Model with Focused Semantic Components

Designing a semantically rich semantic model is essential for enabling AI agents to interpret and use data effectively. The semantic layer should not only represent business logic accurately, but also anticipate how agents will consume and reason over it. Agents are aware of the relative semantic strength of different model components. As a result, they prioritize interpreting and using metrics first, followed by calculated fields, and lastly, raw fields. This prioritization reflects the assumption that curated metrics and calculated logic carry higher intent and clarity.

The following best practices enrichments will support the Agents readability of a Semantic Model:

## Metrics as a Semantic Anchor

A metric is a high-value semantic object designed to answer specific business questions with clarity. It combines:

- Targeted calculations
- Time-scoped logic
- Contextual filtering
- Business-consistent formatting

> **Best Practice**: Define a metric for each high-impact and highly used calculated field or field measure. This guides AI agents in understanding where and when specific logic applies.

## Predefined Calculated Fields

Calculated fields should encapsulate business logic that agents would otherwise need to infer. Defining them in advance improves answer quality, reduces ambiguity, and boosts trust.

> **Best Practice**: Only include calculated fields with clear analytical purpose. Avoid overloading the model with rarely-used or overly niche calculations.

## Explicit Field Type

Classifying fields as measures or dimensions at the metadata level enhances agent reasoning. It signals how data should behave in analytical workflows (for example aggregate vs. filter).

> **Best Practice:** Add semantic roles (Field Type) to all fields. These cues help agents apply them correctly in queries.

## Structuring the Data Model for Agent Confidence

Although island tables or isolated objects are sometimes intentionally created within a semantic model for distinct use cases, they can cause confusion. Both agents and users may misinterpret these structures or try to query them as if they are connected, even though relationships are not defined.

> **Best Practice**: Make sure all objects are part of a single connected cluster. Avoid ambiguous or unjoined objects that could trigger invalid responses from agents.

# Conclusion

Building AI-ready semantic models is an investment that pays dividends in user trust, agent performance, and business agility. By following these best practices, you not only future-proof your data models – you also accelerate your journey to scalable, agent-driven analytics.

# AI Readiness Implementation Checklist

Use this checklist when implementing the best practices outlined in this article to validate and improve your semantic model:

**Labeling & Descriptions**

- [ ] Semantic Model Description is specific and purposeful
- [ ] All field labels are specific and contextual (no “ID”, “Name” without qualifiers).
- [ ] Descriptions are provided for all custom objects and fields.
- [ ] Descriptions for standard Salesforce fields are left unchanged or enhanced without contradicting known meaning.

**Ambiguity Management**

- [ ] Similar fields/metrics are differentiated in name, description, and purpose.
- [ ] Calculated fields and regular fields with similar logic are reviewed for overlap.

**Calculated Fields & Metrics**

- [ ] Only necessary and interpretable calculated fields are included.
- [ ] Metrics differing only by filters are clarified or consolidated.
- [ ] Each metric/field has a clearly defined analytical purpose.

**Semantic Enrichment**

- [ ] Metrics are defined for high-impact KPIs.
- [ ] Matrices are used to encapsulate time/context logic and formatting.
- [ ] Field types (measure/dimension) are explicitly set for each field.

**Model Structure**

- [ ] All tables are connected in a single graph to avoid misinterpretation.